

Practical Implementation of the Kelly Criterion

Reproduction and Statistical Reassessment of Kelly Literature

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1 Introduction

The Kelly Criterion (Kelly 1956) prescribes the growth-optimal allocation $f^* = \Sigma^{-1}(\mu - r_f \mathbf{1})$ but is sensitive to estimation noise, the mean-vs-median divergence of long-horizon log-wealth, and deployment-side constraints (Breiman 1961; Thorp 2011). Carta and Conversano (2020) — hereafter CC20 — extend Kelly to multi-asset equity portfolios via a Taylor expansion of $\mathbb{E}[\log W]$ and report that aggressive Kelly multiples deliver superior out-of-sample (OOS) performance on EuroStoxx 50 versus tangent, MinVar, and equal-weight benchmarks. The claim is taken at face value: no confidence intervals, no hypothesis tests.

This study has four objectives. *(O1)* Validate Kelly geometry under known GBM parameters via Monte Carlo across four horizons. *(O2)* Replicate CC20’s pipeline — single-equity Banca Intesa, in-sample EuroStoxx 50 multi-asset Kelly, and the long-only OOS rolling-window backtest. *(O3)* Attach formal inference via HAC (Newey-West) tests on pairwise daily return differences. *(O4)* Probe robustness by lifting long-only constraints and repeating on Dow Jones Industrial Average (DJI).

The central finding is that CC20’s ranking is not robust. The long-only budget cap mechanically collapses Full, Double, and Triple Kelly onto an *identical* portfolio, so the apparent dominance of aggressive multiples is an optimizer artifact. HAC tests confirm it: only Half-vs-Full reaches significance ($p = 0.014$), with a 4 bp/day gap driven by the collapse. Lifting the cap separates the variants but degrades all of them (Sharpe 0.10–0.18, drawdowns 72–97%), tracking the volatility-pumping geometry the Monte Carlo predicts. On DJI, long-only Kelly is dominated by Buy-and-Hold on drawdown and all Kelly variants cluster tightly on Sharpe.

2 Data and Methodology

We use 35 surviving EuroStoxx 50 constituents with daily adjusted closes from Yahoo Finance (`tidyquant`, `quantmod`), January 2000 through December 2018, after a 20%-missing filter, forward-fill, and complete-case screening. Banca Intesa (ISP.MI) covers 2007–2018 daily; the DJI extension uses $\hat{\text{DJI}}$ over the same window. Monte Carlo paths are GBM with $\mu_{\text{ann}} = 0.12$, $\sigma_{\text{ann}} = 0.40$, $r_f = 0.01$ (theoretical full Kelly ≈ 0.6875); wealth evolves as $W_{t+1} = W_t[(1 - f)e^{r_f \Delta t} + f e^{r_t}]$.

The multi-asset Kelly QP follows CC20’s Taylor expansion of $\mathbb{E}[\log(1 + \mathbf{w}^\top \mathbf{r})]$:

$$\max_{\mathbf{w}} \mathbf{w}^\top (\mu - r_f \mathbf{1}) - \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = f, w_i \geq 0,$$

with $f \in \{0.5, 1, 2, 3\}$ for Half / Full / Double / Triple Kelly. Long-only clips f to $[0, 1]$; unconstrained allows shorts with leverage capped at $|f| \leq 3$. The QP is solved via `quadprog::solve.QP` after a `Matrix::nearPD` projection. Tangent and MinVar use the same machinery with $\mathbf{1}^\top \mathbf{w} = 1$. The OOS protocol is a 756-day rolling window with monthly re-optimization (`portfolioBacktest`, `lookback=252*3`, `optimize_every=rebalance_every=21`). For inference, daily return-differences $d_t = r_t^A - r_t^B$ are regressed on a constant with Newey-West HAC standard errors at five lags via `sandwich::NeweyWest` and `lmtest::coeftest`, testing $H_0 : \mathbb{E}[d_t] = 0$.

3 Results

3.1 Monte Carlo validation under known parameters

The four-horizon Monte Carlo isolates Kelly geometry from estimation noise. At 10,000 trades — roughly 40 years of daily trading — the divergence between mean and median terminal wealth across

multiples is severe (Table 1; sample wealth trajectories and full distributional shape in Appendix Figures A1 and A2). Triple Kelly’s mean of 518.5 sits on top of a 0.01 median and 79% probability of loss; Double Kelly posts a $184\times$ mean against a $1.67\times$ median; Full Kelly has the highest median (6.89) at 12% loss probability; only Half Kelly is unambiguous (4% loss probability, median 4.54).

Table 1: Monte Carlo terminal wealth at 10,000 trades (1,000 paths).

Strategy	Mean	Median	SD	P(Loss)	P(W>2)	P(W>10)
Half Kelly	7.03	4.54	9	4.0%	83.6%	20.0%
Full Kelly	31.47	6.89	94	12.3%	76.3%	41.7%
Double Kelly	183.80	1.67	1803	44.5%	48.3%	29.5%
Triple Kelly	518.50	0.01	11867	79.0%	17.5%	10.8%

Under log-normal wealth, $\mathbb{E}[\log W] < \log \mathbb{E}[W]$ by Jensen’s inequality and the gap widens with $f^2\sigma^2$. Aggressive multiples buy a larger mean by accepting a much smaller median: under known GBM parameters, Triple Kelly is not aggressive but actively destructive of typical investor outcomes. Any empirical case for Triple Kelly must clear a bar the noiseless simulation already shows it cannot.

3.2 Single-equity replication: Banca Intesa, 2007–2018

The Banca Intesa backtest (ISP.MI, daily, 2007–2018) reproduces CC20’s published single-asset ranking: Full Kelly delivers the highest terminal wealth (1.193, CAGR +1.53%) at -23.4% max drawdown; Half Kelly trails marginally (1.176, +1.40%) at -11.1% drawdown; Triple Kelly underperforms (-65.0% drawdown, terminal 0.937); and all three Kelly variants dominate Buy-and-Hold (0.619, -81.3%). Cumulative wealth paths are shown in Appendix Figure A3. The result is unsurprising — the static-fraction backtest amounts to a constant-rebalanced portfolio whose advantage over Buy-and-Hold under volatile single-asset paths is a textbook result (Cover 1991) — and the interest is whether multi-asset Kelly survives the OOS pipeline below.

3.3 In-sample EuroStoxx 50, 2000–2018

The in-sample multi-asset Kelly portfolio loads 62.3% on ITX.MC and 37.7% on VIV.PA — a two-asset corner solution typical of unconstrained Kelly under sample-mean inputs — and posts 19.07% annual return at 21.14% annual volatility, against tangent (16.41%/16.45%), MinVar (7.81%/12.25%), and equal-weight (7.80%/17.04%). The cumulative-wealth trajectory ends near $19\times$ initial capital against $\sim 12\times$ for tangent and $\sim 3\times$ for the benchmarks; the mean-variance plane with all four optimal portfolios is shown in Appendix Figure A4. The dominance is real but in-sample: the same sample mean and covariance the optimizer maximizes against drive the realized returns, so the comparison is not predictive — it is the OOS sections below that test whether it survives.

3.4 Out-of-sample EuroStoxx 50: long-only

Once we impose the long-only budget cap $\sum w_i = f \leq 1$, the constraint binds for any $f \geq 1$ and Full, Double, and Triple Kelly *collapse onto an identical portfolio* — same QP solution, same daily returns, same metrics. Table 2 makes the collapse visible: Full / Double / Triple Kelly produce identical Sharpe (0.621), max drawdown (0.639), and annual return (16.87%); only Half Kelly differs (Sharpe 0.556, return 7.87%, drawdown -35.2%); MinVar yields Sharpe 0.480.

Table 2: Long-only OOS performance, EuroStoxx 50, 2005-2018. Full / Double / Triple Kelly are identical: the budget constraint binds for $f \geq 1$.

Strategy	Sharpe	Max DD	Ann Return	Ann Vol	Sortino	CVaR (0.95)
Half Kelly	0.556	0.352	7.87%	14.15%	0.835	0.0195
Full Kelly	0.621	0.639	16.87%	27.17%	0.925	0.0380
Double Kelly	0.621	0.639	16.87%	27.17%	0.925	0.0380
Triple Kelly	0.621	0.639	16.87%	27.17%	0.925	0.0380
MinVar	0.480	0.580	8.55%	17.79%	0.685	0.0261

A naive reading endorses CC20: aggressive Kelly variants beat MinVar on Sharpe, return, and Sortino, with the collapsed-aggressive group reaching $\sim 6\times$ by 2018 against $\sim 2.5\times$ for Half Kelly and MinVar (Appendix Figure A5). Inference reverses the reading.

3.5 Statistical inference: HAC pairwise tests

Newey-West HAC tests (5 lags) on daily return differences $d_t = r_t^A - r_t^B$ deflate the apparent gaps. Only Half-vs-Full reaches conventional significance ($p = 0.014$), and even there the mean difference is just 4 bp/day. Half-vs-MinVar is essentially zero ($p = 0.86$); Full-vs-MinVar is +3 bp/day with $p = 0.14$. Pairwise mean differences with 95% HAC confidence intervals are visualized in Appendix Figure A6.

Table 3: HAC (Newey-West, 5 lags) tests on daily return differences, long-only OOS.

Pair	Mean Diff	HAC SE	t-stat	p-value
Half Kelly vs Full Kelly	-0.0004	0.0001	-2.4646	0.0138
Half Kelly vs MinVar	0.0000	0.0002	-0.1785	0.8584
Full Kelly vs MinVar	0.0003	0.0002	1.4736	0.1407

The Half-vs-Full significance is mechanically driven by the constraint collapse: Full Kelly’s “outperformance” is the difference between an $f = 0.5$ and an $f = 1$ corner solution under the same QP. Half-vs-MinVar and Full-vs-MinVar — the comparisons CC20’s narrative most directly relies on — both fail to reject. *Once realistic long-only constraints are imposed, the choice of Kelly multiplier has no statistically distinguishable effect on out-of-sample returns.* With $\sim 3,300$ daily observations across 13 years, this is the longest sample CC20’s data permits, and the failures to reject are therefore informative rather than data-starved. The absence of any such inferential check in CC20 is the structural weakness our extension targets.

3.6 Out-of-sample EuroStoxx 50: unconstrained

Lifting the budget cap separates the four Kelly variants but degrades all of them. Sharpe ratios fall to 0.10–0.18; max drawdowns climb to 72% (Half), 85% (Full), 96% (Double), 97% (Triple); annual returns are flat across multiples (5.5–7.5%) while volatility scales linearly with f from 31% to 62%. Appendix Figure A7 shows the Half Kelly allocation stack: period-long short positions in 2009–2010 and leverage hitting the $|f| = 3$ cap repeatedly. This is the over-leverage pathology the Monte Carlo predicts under known parameters: the long-only constraint is doing real work in the constrained backtest by forcing the optimizer away from Kelly geometry.

3.7 Generalization: Dow Jones Industrial Average

On the DJI single-index backtest (2010–2018), Buy-and-Hold posts a -20% max drawdown; Half and Full Kelly Long-Only also hold -20% ; the unconstrained Half through Triple Kelly variants run to -54% , -54% , -58% , and -65% . Sharpe ratios cluster tightly across all variants and Buy-and-Hold sits near the top of the range. The CC20 ranking does not generalize; the result that does is the conservative one — Half/Full Kelly Long-Only matches Buy-and-Hold on drawdown.

4 Discussion and Conclusion

Three findings emerge that CC20 do not report. (*F1*) The Monte Carlo establishes that Kelly’s mean-vs-median divergence is structural, not estimation-driven: aggressive multiples destroy median wealth even under known parameters. (*F2*) The long-only OOS budget cap mechanically collapses Full / Double / Triple Kelly onto an identical portfolio, so CC20’s headline ranking is evidence about a constraint, not about Kelly fractions. Lifting the cap separates the variants but degrades all of them in line with the Monte Carlo prediction — the consistency check confirming the pathology is geometry. (*F3*) HAC inference shows the only significant long-only pairwise gap is the constraint-collapse artifact; Full-vs-MinVar and Half-vs-MinVar — the comparisons CC20’s narrative most directly relies on — fail to reject on the longest sample CC20’s data permits.

Backtest comparisons are inferential exercises, and the absence of formal hypothesis tests in much of the practitioner literature lets cumulative-wealth panels do work they cannot honestly support. Half Kelly is the only fraction with a defensible record across all four diagnostics — Monte Carlo median wealth, single-equity drawdown, OOS long-only Sharpe, and DJI generalization — and the practical recommendation is the conservative one. Future work should attach formal inference to existing Kelly comparisons, explore μ/Σ shrinkage under rolling-window OOS, and test the constraint-collapse mechanism on additional equity panels.

5 Limitations

Several limitations remain. Kelly optimization is highly sensitive to estimation error in expected returns and covariance matrices, particularly in rolling-window out-of-sample settings where small input changes can materially affect portfolio weights. Besides, the analysis also assumes relatively stable return dynamics. But real financial markets experience regime shifts, crises, structural breaks that may violate these assumptions. In addition, the backtests do not incorporate transaction costs, slippage, other market frictions. This could materially reduce the performance of highly leveraged or frequently rebalanced strategies. Finally, long-only portfolio constraints alter the theoretical Kelly optimum and may mechanically affect portfolio comparisons across Kelly multiples.

6 References

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